



CamTech

— KNOWLEDGE, REASON, CHARACTER —

Cambodia University of Technology and Science

[Your Full Name]

A Thesis Submitted in Partial Fulfillment of the Requirements for
the Degree of

Bachelor Degree of [Program Name]

Cambodia University of Technology and Science
Faculty of Engineering

YOUR THESIS TITLE HERE

By

Your Full Name

(ID: Your Student ID)

Supervisor : [Supervisor's Name and Title]

Mentor : [Mentor's Name and Title]

[Month, Year]

Phnom Penh, Cambodia

Declaration of Originality

I, **[Your Full Name]**, hereby declare that this thesis, titled "**[Your Thesis Title Here]**", submitted in partial fulfillment of the requirements for the degree of Bachelor of **[Degree Type]** at the Cambodia University of Technology and Science, is my own original work. This work was carried out during my industrial placement at **[Company Name]** from **[Start Date]** to **[End Date]**. All sources and references used were properly acknowledged and cited.

Full name : [eg. Kim Lorrentz]

Date : [eg. June 25, 2025]

Report Approval

This report, titled "**Your Report Title Here**", submitted by **Your Full Name** (ID: **Your Student ID**), has been approved by the following members of the examination committee as a partial fulfillment of the requirements for the degree of **Bachelor degree in [Degree Type]** at **[Your Program Name]** at the Cambodia University of Technology and Science.

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Acknowledgements

Start your acknowledgements here. Thank your supervisor(s), committee members, funding bodies, colleagues, friends, and family. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

Abstract

For an undergraduate thesis abstract, write one paragraph (150–250 words) that briefly summarizes your project. Begin by stating the background or problem you are addressing and the objective of your study. Then, describe the methods you used to carry out the research. Next, summarize the main results or findings. Finally, conclude by explaining the significance of your results or the potential application of your work. Keep the writing clear and simple, avoid detailed explanations or references, and use past tense for work you have completed.

Keywords:,....

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Chapter 1

Introduction

1.1 Background of the Study

In this section, provide a general introduction to your research topic. Describe the larger field or industry you are working in, its significance, and why the topic is important. For example, if you're working on robotics or automation, briefly explain how these technologies are transforming industries like manufacturing, construction, or healthcare.

1.2 Company and Team Overview

Introduce the company or organization where you conducted your industrial placement. Provide a brief history of the company, its mission, and key areas of focus. Then, describe the team or department you worked with, including its goals and how your project fits within the organization's objectives. For instance, if you were working with a robotics team, you can explain how your project contributed to the development of new automation systems.

1.3 Project Overview

Describe the specific project you worked on during your industrial placement. Include the project's objectives, scope, and any technical challenges you faced. Explain how your project aligns with the company's broader goals and discuss any expected outcomes or

deliverables. For example, your project might involve designing or improving a vision-based welding robot for construction tasks.

1.4 Research Objectives (if any)

Clearly define the main objectives of your research or project. What are you trying to accomplish, and what key questions are you attempting to answer? For example, your objective might be to improve the accuracy and efficiency of robotic welding by integrating vision systems into the process.

1.5 Thesis Structure

Briefly outline the structure of the thesis, providing a summary of each chapter. For example:

- **Chapter 1: Introduction** – This chapter provides background information on the study, the company, and the research project.
- **Chapter 2: Problem Statement** – This chapter identifies the main problem the project aims to solve.
- **Chapter 3: Literature Review** – This chapter reviews relevant literature and existing research related to the topic.
- **Chapter 4: Methodology** – This chapter describes the methods and techniques used in the research.
- **Chapter 5: Results and Discussion** – This chapter presents and analyzes the findings from the research.
- **Chapter 6: Conclusion and Recommendations** – This chapter concludes the study and offers recommendations based on the findings. =

Chapter 2

Problem Statement

2.1 Background of the Problem

In this section, provide a brief background to the problem your research addresses. Describe the broader context and why this issue is significant. For example, you could explain the challenges faced by industries in robotics, automation, or welding, and why they are relevant to your study. Discuss any existing limitations or gaps in current solutions.

Clearly define the specific problem your research aims to solve. For example, you might identify inefficiencies in current welding processes, limitations of existing robotic systems, or challenges with integrating vision-based systems into industrial workflows.

2.2 Impact of the Problem

Explain the impact of this problem on the company, industry, or specific field of study. Discuss how the problem affects productivity, quality, safety, or costs. For example, you could highlight how manual welding is time-consuming and prone to errors, leading to increased operational costs and lower quality.

2.3 Scope of the Problem

Narrow down the scope of the problem to focus on what your project addresses. Explain the specific aspect of the problem that you will tackle in your research. For instance,

your project may focus on automating a specific part of the welding process or improving the accuracy of robotic vision systems for better alignment.

2.4 Summary of the Problem

Summarize the problem in one or two sentences to clearly state what your research aims to solve. This will set the stage for the objectives and methodology you will describe later in your thesis.

Chapter 3

Methodology

...

[To be changed according to your project]

Referencing other chapters

??, 3, 4, 5 and 6

...

Chapter 4

Implementation

...

[To be changed according to your project]

...

Chapter 5

Evaluation and Discussion

...

Chapter 6

Conclusion

6.1 Conclusion

In this section, summarize the key findings and outcomes of your project. Reflect on the objectives you set at the beginning of the project and discuss whether or not they were achieved. Highlight any significant results, discoveries, or conclusions drawn from your project. For example, you might conclude by stating how your work improved the robotic welding process, enhanced accuracy, or contributed to industrial automation. Also, discuss any unexpected results or challenges you faced during the project.

6.2 Contributions of the Project

Describe the main contributions your project has made to the field. This could include novel methodologies, advancements in technology, or new insights into the application of robotics or automation. Explain how your project adds value to the company, the industry, or the academic community.

6.3 Limitations of the Project

Acknowledge any limitations or constraints in your project. This might include things like limited resources, time constraints, or challenges in data collection or analysis. Being

transparent about these limitations shows that you have a realistic understanding of the scope of your work.

6.4 Future Work

In this section, discuss potential directions for future research or improvements in the field. Suggest how your work can be expanded, refined, or applied in new areas. For example, you could propose enhancing the vision-based system, improving automation processes in other industries, or exploring new technologies that could complement your project. Suggest specific experiments or studies that could build on your findings and push the work further.

...

References

- [1] A. A. Aardvark, “Article title,” *Journal One*, vol. 1, no. 1, pp. 1–8, 1900.
- [2] A. A. Abramson and B. B. Barbie, “Article title,” *Journal Two*, vol. 1, no. 1, pp. 9–17, 1900.
- [3] A. A. Abramson, B. B. Barbie, and C. C. Rider, “Article title,” *Journal Three*, vol. 1, no. 1, pp. 192–244, 1900.

Appendix A

Extra Stuff

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

A.1 Spacing & Type

This is a section. This is a citation without brackets. and this is one with brackets [1]. Multiple [1], [2], [3] Here’s a reference to a subsection: A.1.1. The body of the text and abstract must be double-spaced except for footnotes or long quotations. Fonts such as Times Roman, Bookman, New Century Schoolbook, Garamond, Palatine, and Courier are acceptable and commonly found on most computers. The same type must be used throughout the body of the text. The font size must be 10 point or larger and footnotes¹ must be two sizes smaller than the text² but no smaller than eight points. Chapter, section, or other headings should be of a consistent font and size throughout the ETD, as should labels for illustrations, charts, and figures.

¹This is a footnote.

²This is another footnote.

A.1.1 Creating a Subsection

Creating a Subsubsection

This is a heading level below subsubsection And this is a quote:

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

This is a table:

Table A.1: This is a caption.

A	B
a1	b1
a2	b2
a3	b3
a4	b4

The package “upgreek” allows us to use non-italicized lower-case greek letters. See for yourself: β , $\boldsymbol{\beta}$, β , $\boldsymbol{\beta}$. Next is a numbered equation:

$$\|\mathbf{X}\|_{2,1} = \underbrace{\sum_{j=1}^n f_j(\mathbf{X})}_{\text{convex}} = \sum_{j=1}^n \|\mathbf{X}_{:,j}\|_2 \quad (\text{A.1})$$

The reference to equation (A.1) is clickable.

A.2 Figure and Graphic

The experiment as shown in Fig.A.1, demonstrates a long range of motion with a high conditional number. In this trajectory, the highest conditional number hits up to 2900. However, the result shows that the state error and controller cost are relatively small as shown in table A.1.

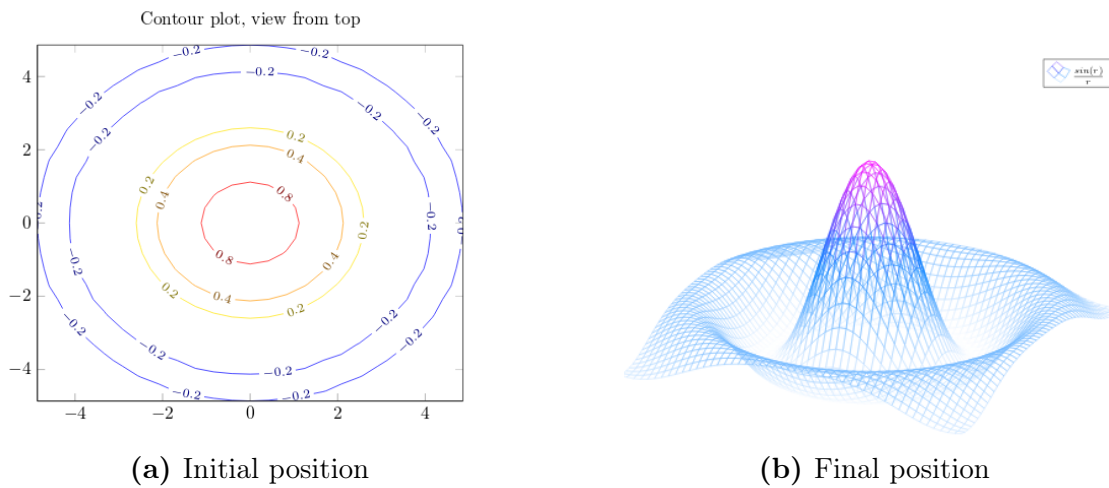


Figure A.1: Long range motion: line trajectory

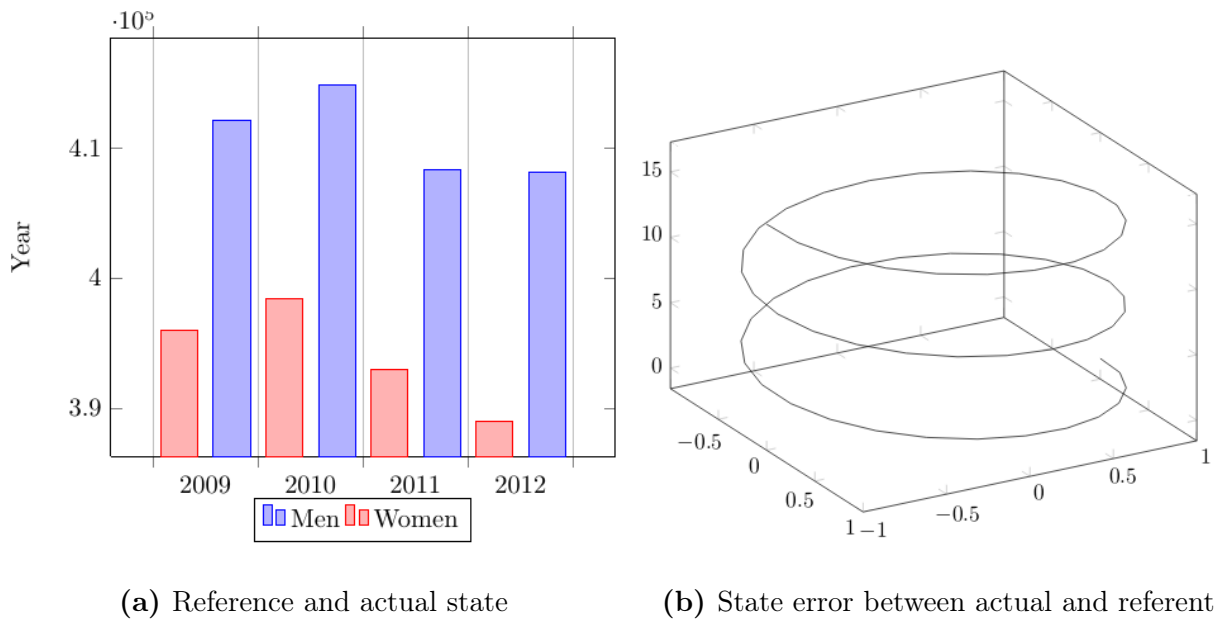
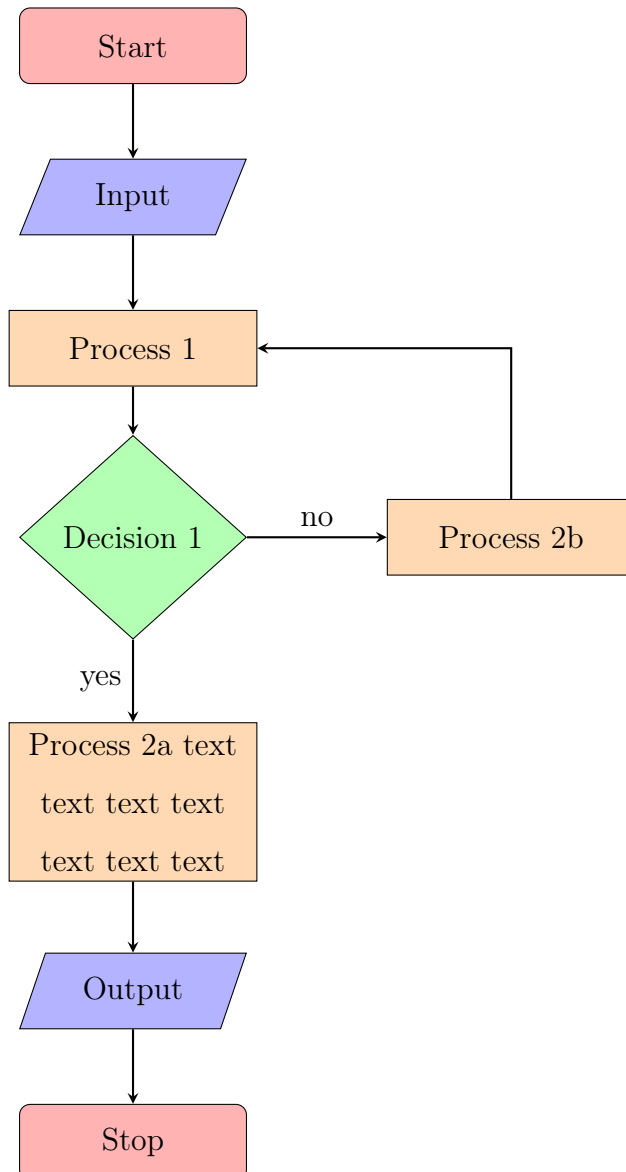


Figure A.2: State tracking of long range motion(line trajectory)

A.3 Flowchart

The following example contains the code and can be opened directly in Overleaf. This code adds the option `trapezium stretches=true` to the node style definition, helping to ensure trapezium nodes are the same width.



A.4 Algorithm and pseudo code

Here we demonstrate how algorithms or pseudocode can be typeset using the algorithmic environment provided by the `algpseudocode` package.

Algorithm 1 Example While Loop

```
1:  $i \leftarrow 0$ 
2: while  $i < 10$  do
3:   Print  $i$ 
4:    $i \leftarrow i + 1$  ▷ This is a comment
5: end while
```

A.5 Code Display

Example of display of your code.

```
# This is a black comment
def hello():
    print("Hello , world!") # Keyword 'def' is blue , comment is black
```

A.6 Theorems, Corollaries, Lemmas, Proofs, Remarks, Definitions, and Examples

Theorem 1. *Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.*

Proof. I’m a (very short) proof. □

Lemma 1. *I’m a lemma.*

Corollary 1. *I include a reference to Thm. 1.*

Proposition 1. *I’m a proposition.*

Remark. I’m a remark.

Appendix B

Even More Extra Stuff

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.